



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

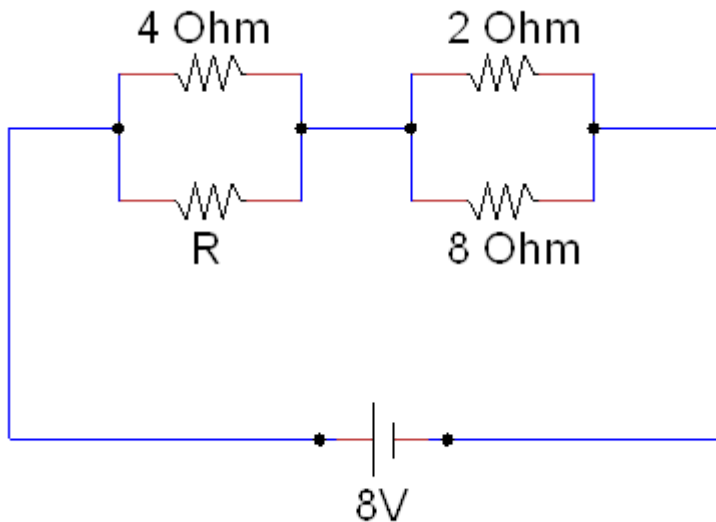
END SEMESTER ASSESSMENT (ESA) - JAN 2023

UE23EE141A - Elements of Electrical Engineering

Total Marks : 100.0

1.a. The total power consumed by the network shown in the figure is 16 W. Find the value of R and total current

(5.0 Marks)

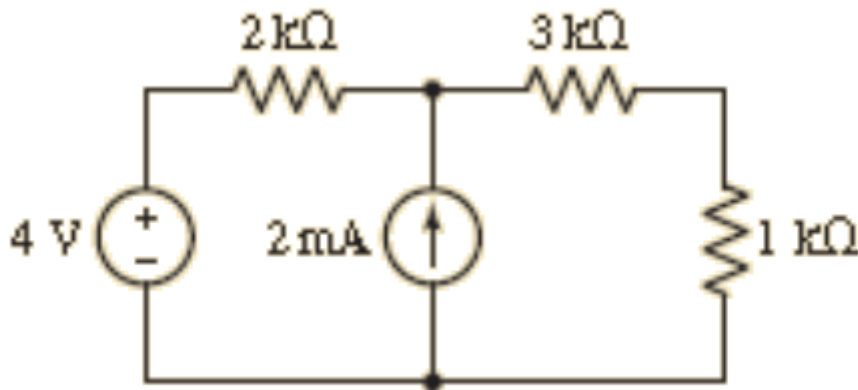


1.b. With proper nomenclature, derive how a delta connected network of resistors can be transformed to its equivalent star network . (6.0 Marks)

1.c. (i) State Thevenin's theorem

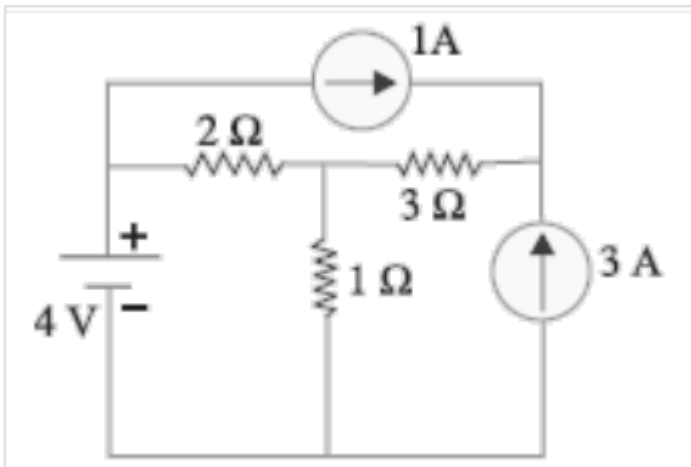
(ii) Find the Thevenin's equivalent circuit for the network shown with $1\text{ k}\Omega$ resistor as load

(7.0 Marks)



1.d. Find current through 1Ω resistor using SUPERPOSITION theorem for the circuit shown

(7.0 Marks)

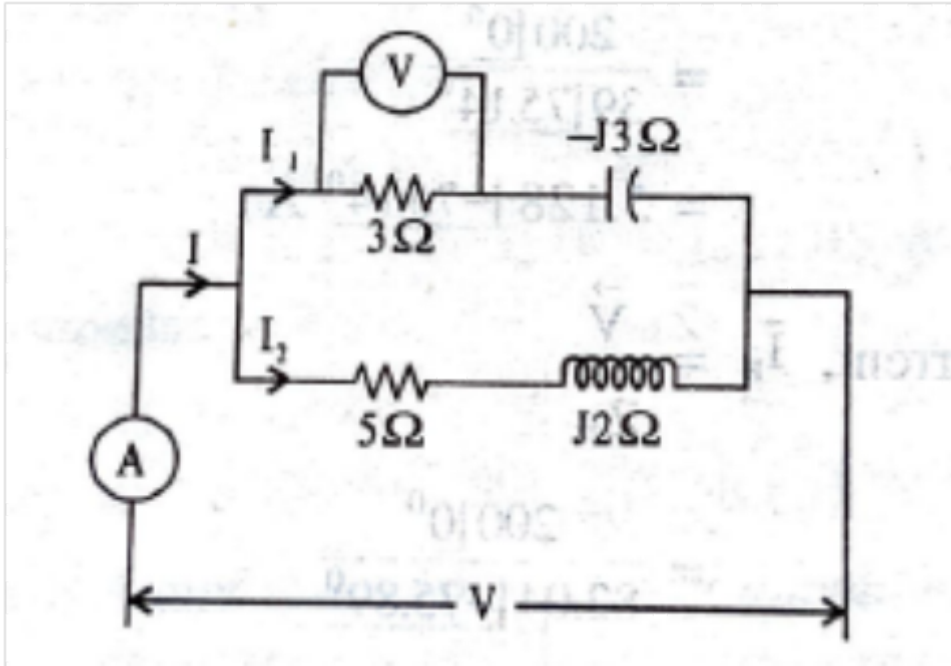


2.a. A coil of resistance of 10Ω and an inductance of 0.1 H is connected in series with a $150\mu\text{ F}$ capacitor across a 200 V , 50 Hz supply. Calculate Impedance, current, power factor, voltage across coil and voltage across capacitor

(7.0 Marks)

2.b. Find the reading of the ammeter when the voltmeter across $3\ \Omega$ resistor in the circuit shown reads 45V

(5.0 Marks)



2.c. A single phase motor connected to 400 V, 50 Hz supply takes 31.7A at a power factor of 0.7 lagging. Calculate the capacitance required in parallel with the motor to raise the power factor to 0.9 lagging (7.0 Marks)

2.d. The Voltage and current for a circuit with two elements in series are expressed as follows :

$$v(t) = 170 \sin(6280t + \pi/3) \text{ V}, i(t) = 8.5 \sin(6280t + \pi/2) \text{ A}$$

- (i) Plot the two waveforms & Vector diagram
- (ii) Determine the frequency in Hz
- (iii) Determine the power factor stating its nature
- (iv) What are the values of the elements?

(6.0 Marks)

3.a. When the three identical star-connected coils are supplied with 440 V, 50 Hz, 3 phase supply. A 1-phase wattmeter whose current coil is connected in line R and pressure coil across the phase R and neutral reads 6 kW and the ammeter connected in R-phase reads 30 Amp. Assuming RYB phase sequence find

- (i) Resistance and reactance of the coil
- (ii) the power factor of the load
- (iii) Reactive power of 3- ϕ load.

(6.0 Marks)

3.b. A 3 phase delta connected load consumes a power of 100KW taking a lagging current of 200 A at a line voltage of 400 V, 50 Hz. Find the parameters of each phase. What would be the power consumed if the load were connected in star?

(6.0 Marks)

3.c. Following are the loads for a typical household

Sl.No.	Name of the appliance	Wattage	Average consumption hours per day
1	Four ,Tube lights	60 Watt per Tube light	10 hours each
2	Four , Table fans	75 Watt per Fan	12 hours each
3	One Room Heater	1KW	1 hour
4	Refrigerator	100 Watt	16 hours
5	Television	50 Watt	10 hours
6	Iron box	750 Watt	1 hour

Considering 30 days a month

Determine

(i) The total number of units consumed in a month

(ii) Monthly bill for the above consumption units considering a domestic connection of 3KW sanctioned load.

Neglect any other charges

Tariff Details

Sl.No.	Types of Charges	Tariff details
1	Fixed Charges for sanctioned load	Rs. 50 for first KW Rs. 60 for every additional KW
2	Energy Consumption Charges	0-30 units ---- Rs.3.50 per unit 31-100 units --- Rs.4.95 per unit 101-200 units --- Rs.6.50 per unit 201-300 units --- Rs.7.55 per unit 301-400 units --- Rs.7.60 per unit Above 400 units--Rs.7.65 per unit

(7.0 Marks)

3.d. (i) What is Earthing ? Why earthing is necessary ?

(ii) Explain the working principle of a Miniature Circuit Breaker (6.0 Marks)

4.a. (i) A 4 pole dc shunt motor takes 22.5 A from a 250V supply . The armature resistance is 0.5Ω and field resistance is 125Ω . The armature is wave wound with 300 conductors. If the flux per pole is 0.02 wb. Calulate back emf, speed, torque developed and power developed .

(ii) What is Back EMF ? What is the value of Back EMF at the point of starting of the DC motor? (7.0 Marks)

4.b. (i) Explain the concept of Rotating Magnetic Field in a 3 phase induction motor.

(ii) A 12 pole 3 phase alternator is coupled to an engine running at 500 RPM. It supplies an induction motor which has full load speed of 1440 RPM. Find %slip and number of poles of the motor. (7.0 Marks)

4.c. (i) The primary winding of a single phase transformer is connected to a 240 V ,50 Hz supply . The secondary winding has 1500 turns . If the maximum value of the core flux is 0.002wb, determine number of turns on primary winding ,secondary induced voltage & net cross sectional area if the flux density has maximum value of 0.465 Tesla.

(ii) A 5 kVA , 50 Hz, single phase transformer has primary and secondary turns 120 & 80 respectively . At a certain flux density the induced emf per turn in primary is 2.5 V. Determine primary and secondary voltages. Also calculate current in both the windings. (6.0 Marks)

4.d. Derive EMF equation of a DC generator.

(5.0 Marks)



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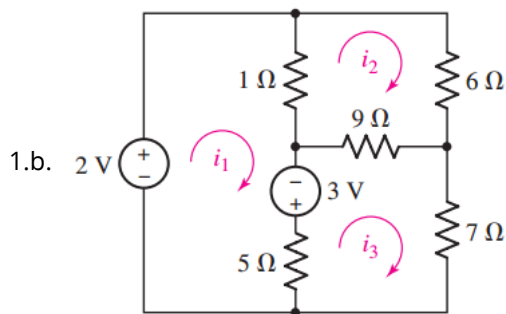
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END SEMESTER ASSESSMENT (ESA) - Jan-Feb

UE24EE141A - Elements of Electrical Engineering

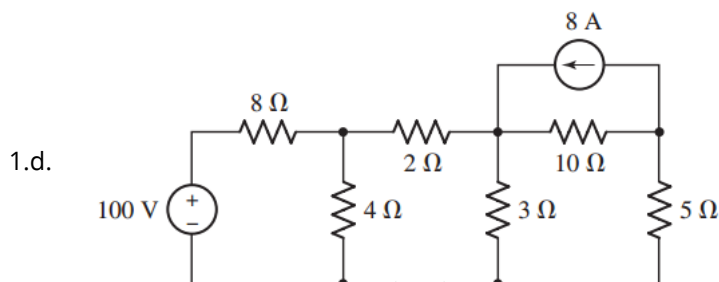
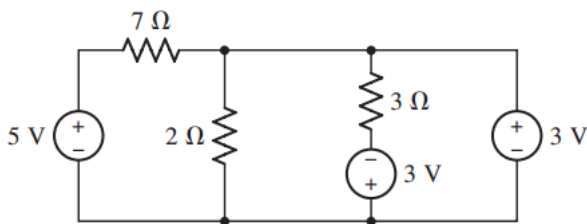
Total Marks : 100.0

1.a. With proper nomenclature, derive how three resistors connected in delta can be transformed to equivalent star. (5.0 Marks)



Considering mesh currents i_1 , i_2 & i_3 as indicated, apply mesh analysis and find them. (6.0 Marks)

1.c. Using Superposition Theorem, determine current through 7Ω resistor in the given network. (Note: Do not apply source transformation) (6.0 Marks)



In the given network, considering 2Ω resistor as load, determine
i) Thevenin's voltage
ii) Thevenin's Resistance

Represent the Thevenin's equivalent for the given network and determine the load current. (8.0 Marks)

2.a. A series RLC network connected across $100\sin(100\pi t)$ Volts draws a current $10\cos(100\pi t - 60^\circ)$ Amperes. Determine

- i) Resistance in the network
- ii) Powerfactor of the network
- iii) Capacitance if inductance is 10mH

Also determine the extra inductance to be added in series to bring the network into resonance. (6.0 Marks)

2.b. i) Write a short note on the advantages of power factor improvement.

ii) A heating load of 5kW, a motor load of 10kVA at 0.8 lagging power factor & another load of 12kW at 0.6 lagging power factor are connected in parallel across a 200V, 50 Hz supply. Determine

- a) Total Active, Reactive and Apparent powers
- b) Powerfactor of the system
- c) Capacitance to be added in parallel to bring the power factor to unity.

(9.0 Marks)

2.c. A resistor 2.5Ω , an ideal inductor of inductance 5.305mH and an ideal capacitor of capacitance $954.93\mu F$ are connected in parallel across 10V, 50 Hz supply. Determine

- i) Admittance of each branch
- ii) Input Admittance of the network in both rectangular and polar forms
- iii) Branch currents and supply current in phasor form (polar form)

Considering supply voltage as reference phasor, draw the phasor diagram indicating branch currents and supply current. (10.0 Marks)

3.a. Write a short note on the following:

- i) Fuse
- ii) RCCB

(4.0 Marks)

3.b. Two wattmeters connected in a balanced three phase 400V, 50Hz system indicate 4kW and 12kW respectively. If the load is star type and inductive in nature, determine

- i) Line current
- ii) Load Resistance per phase
- iii) Load Inductance per phase
- iv) Powerfactor of the Load

(6.0 Marks)

3.c. **With a neat circuit diagram and phasor diagram of two wattmeter method, derive an expression for finding powerfactor in a balanced three phase system.** (7.0 Marks)

3.d. The list of loads and average consumption hours per day of a typical household is given below:

S.No.	Name of the Appliance	Wattage	Average consumption hours per day
1.	Geyser	2000W	1 hour
2.	Refrigerator	100W	24 hours
3.	TV	60W	4 hours
4.	Waterpump	750W	1 hour
5.	Lighting Load	100W	6 hours
6.	Fan Load	200W	3 hours

Considering a 30 day month, Determine

i) Total number of units consumed in a month

ii) Monthly bill for the above consumption units considering a domestic connection of 5 kW sanctioned load with the tariff details given below:

S.No	Type of Charges	Tariff Details
1.	Fixed Charges for the sanctioned load	Rs. 100/- - For first kW Rs. 110/- - For every additional kW
2.	Energy consumption Charges	0 – 50 Units @ Rs. 4.15/- Per Unit 51 – 100 Units @ Rs. 5.6/- Per Unit 101 – 200 Units @ Rs. 7.15/- Per Unit Above 200 Units @ Rs. 8.2/- Per Unit
3.	Fuel Adjustment Charges	@ 12 Paise per unit consumed
4.	Tax applicable only for units consumed	@ 9% (Note: Only applicable for Units consumed and not for Fixed charge adjustment charges)

(8.0 Marks)

4.a. **With proper nomenclature, derive an expression for electromagnetic torque developed by a DC Motor.** (5.0 Marks)

4.b. **A 6 pole, wave wound DC shunt generator having 200 conductors on its armature is supplying power at 200V to the following loads connected in parallel:**

- i) 5kW heating load
- ii) 1kW lighting load

If armature and shunt field resistances are 0.5Ω and 100Ω respectively and it is running at 1500 rpm, determine

- i) Generated EMF
- ii) Electrical power developed
- iii) useful flux per pole

(5.0 Marks)

4.c. i) Give any two basic differences between squirrel cage rotor and phase wound rotor in a three phase induction motor.

ii) A 4 pole, three-phase alternator running at 1800 rpm is supplying power to a three-phase, 6 pole induction motor. Determine

a) Synchronous speed of the induction motor

b) No-load speed if no-load slip is 2%

c) Frequency of rotor currents on full-load if full-load speed is 1140 rpm.

(7.0 Marks)

4.d. A 20kVA, 2000V/400V, 50 Hz single phase transformer with 200 secondary turns has following test data:

OC Test (on LV side) : 400V, 1A , 100W

SC Test (on HV side) : 100V , 10A , 400W

If maximum flux density in the core is 1.4T, determine

i) Number of primary turns

ii) Rated primary and secondary currents

iii) Net cross sectional area of the core

iv) Full load efficiency at 0.6 powerfactor

v) Half-load efficiency at 0.8 powerfactor

vi) overall maximum efficiency

(8.0 Marks)



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END SEMESTER ASSESSMENT (ESA) - Jan-Feb

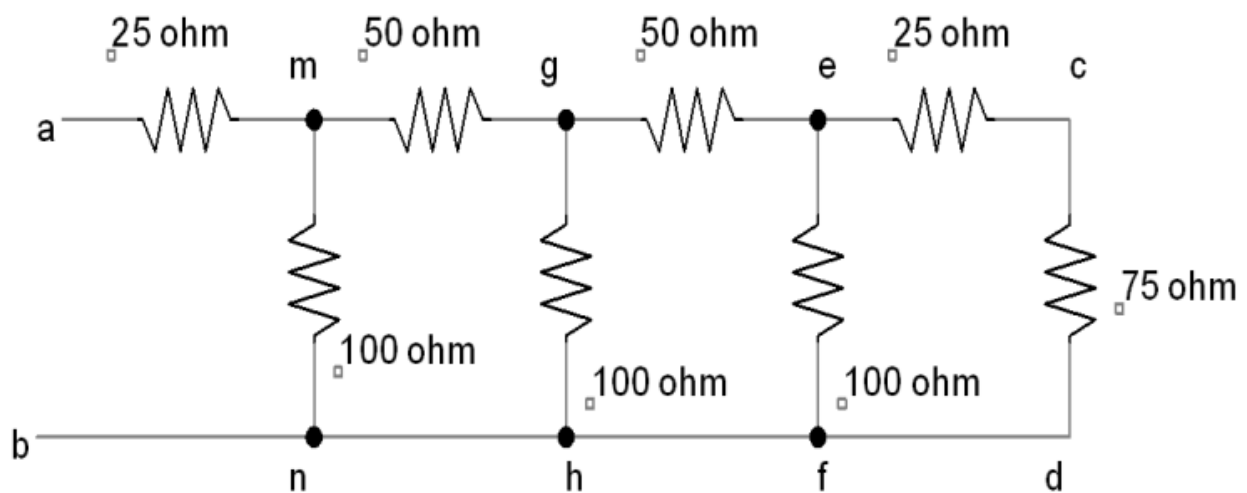
UE24EE141A - Elements of Electrical Engineering

Total Marks : 100.0

1.a. What is the equivalent resistance of the ladder network shown in the figure below

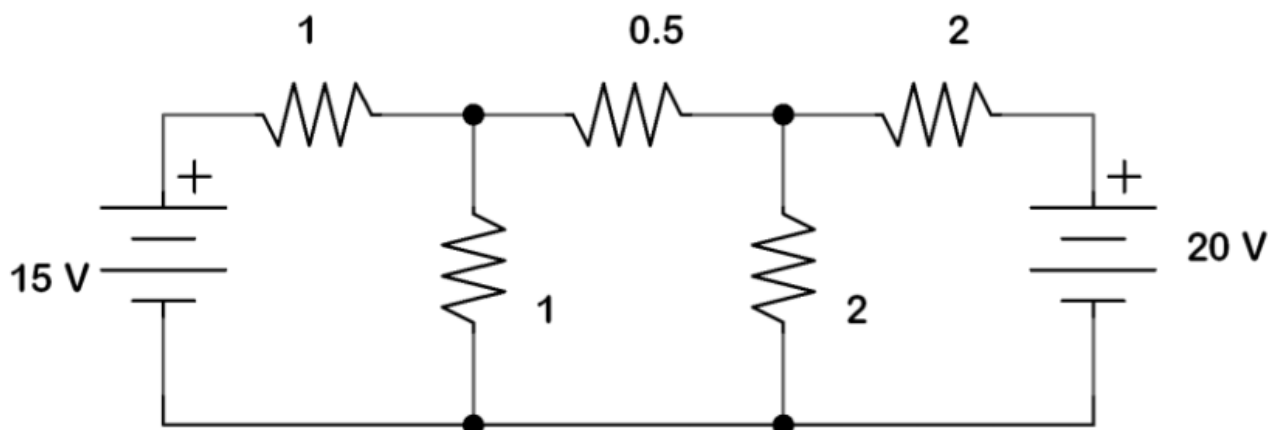
(i) With $75\ \Omega$ load resistor connected as shown

(ii) with $75\ \Omega$ load resistor disconnected?



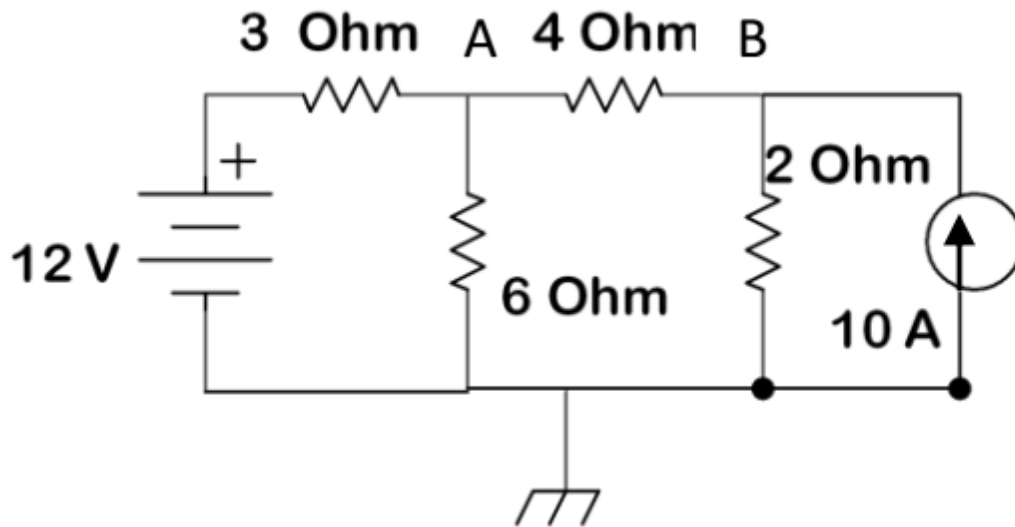
(6.0 Marks)

1.b. Determine current flowing through the resistance $R = 0.5\ \Omega$ in the circuit shown using Superposition Theorem (All resistances in Ω)



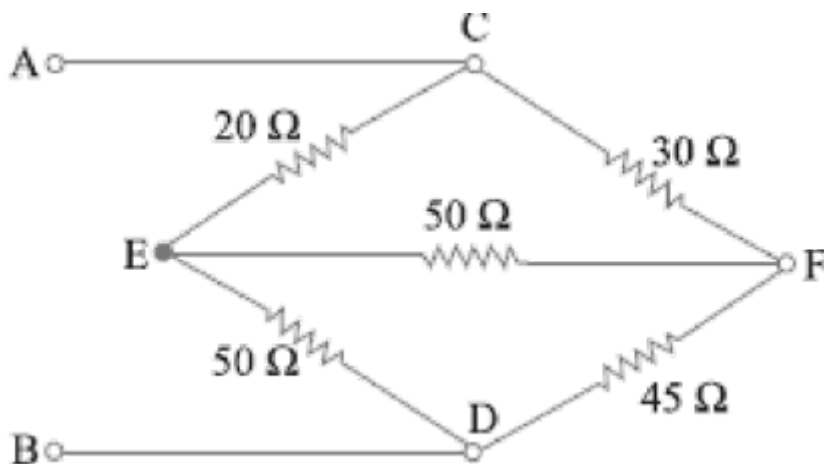
(7.0 Marks)

1.c. Using Thevenin theorem, calculate the current flowing through the 4 Ω resistor



(7.0 Marks)

1.d. Find the resistance at the A-B terminals using Δ -Y transformation.



(5.0 Marks)

2.a. A choke coil takes a current of 2 A, lagging 60° behind the applied voltage of 200V at 50 Hz. Calculate the inductance, resistance and impedance of the coil. Also determine the power consumed when it is connected across 100V, 25Hz supply. (7.0 Marks)

2.b. In a series parallel circuit, the two parallel branches A and B are in series with C. The impedances are $Z_A = 10 + j8 \, \Omega$, $Z_B = 9 - j6 \, \Omega$ and $Z_C = 3 + j2 \, \Omega$. The voltage across C is $(100 + j0) \, \text{V}$. Find the currents I_A and I_B . (6.0 Marks)

2.c. A single-phase motor connected to 400V, 50 Hz supply takes 31.7 A at a power factor of 0.7 lagging. Calculate the capacitance required in parallel with the motor to raise the power factor to 0.9 lagging. (7.0 Marks)

2.d. A resistance of $100 \, \Omega$ is connected in series with $100 \, \mu\text{F}$ capacitor across a 200V, 60 Hz supply. Find impedance, current, power factor, power and voltage across resistance and capacitance. (5.0 Marks)

3.a. A balanced star connected load of 150 kW takes a leading current of 100A with a line voltage of 1100V, 50 Hz. Find the circuit constants of the load per phase. (5.0 Marks)

3.b. The following table gives the number of Units consumed in a typical household

Sl.No.	Name of Appliance	Average Consumption per Day
1	Tube lights	1.2 Unit
2	Fans	1.2 Unit
3	Water Pump	0.75 Unit
4	Refrigerator	2.4 Unit
5	TV	0.5 Unit
6	Grinder	0.3 Unit
7	Geyser	1 Unit
8	Air Conditioner	2 Unit

Consider a 30-days month, Determine

- The total number of units consumed in a month
- Monthly bill for the above consumption units considering a domestic connection of 4kW sanctioned load with the tariff details listed below

Sl.No	Type of charges for sanctioned load	Tariff Details
1.	Fixed charge for sanctioned load	Rs 50 for the 1 st kW Rs 60 for every additional kW
2.	Energy Consumption Charges	0-30 Units Rs 3.50 per unit 31-100 Units Rs 4.95 per unit 101-200 Units Rs 6.50 per unit 201-300 Units Rs 7.55 per unit 301-400 Units Rs 7.60 per unit Above 400 Rs 7.65 per unit

Consider a tax of 8% on total charges.

(6.0 Marks)

3.c. The input power to a 1.6kV, 50 Hz, 3 phase motor is measured by two wattmeter method. The motor is running on full load with an efficiency of 86%. The reading of two wattmeters is 255kW and 85 kW. Determine (a) input power (b) the power factor (c) line current (d) the output power
(6.0 Marks)

3.d. With relevant circuit and vector diagram prove that two wattmeters are sufficient to measure total power in a three phase circuit when the load is star connected. Also show how power factor angle can be found from two wattmeter readings. (8.0 Marks)

4.a. With proper nomenclature derive the EMF equation of a DC generator. (6.0 Marks)

4.b. (i) In a 25 kVA transformer the iron loss and full load copper loss are 350 W and 400 W respectively. Calculate efficiency of transformer at (i) Half load at UPF (ii) $3/4^{\text{th}}$ full load 0.8 pf lag
(ii) A transformer supplies a load of a 32A at 415V. If the primary voltage is 3320 V Find (a) Secondary volt ampere (b) Primary current (c) Primary volt ampere (8.0 Marks)

4.c. A DC motor takes an armature current of 100A at 400 V. The resistance of armature circuit is 0.1Ω . The machine has 6 poles and the armature is lap connected with 864 conductors. The flux per pole is 0.05 Wb. Calculate Speed and the torque developed by the armature .
(5.0 Marks)

4.d. (i) A 6 pole induction motor is supplied by a 8 pole alternator which is driven at 750 RPM. If the induction motor is running at 930 RPM determine its % slip.
(ii) Explain the concept of Rotating magnetic field in a 3-phase induction motor.
(6.0 Marks)



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END SEMESTER ASSESSMENT (ESA) - Feb 2026

UE25EE141A - Elements of Electrical Engineering

Total Marks : 100.0

1.a. Calculate the value of V_{AB} in the circuit shown below , when the switch S is :

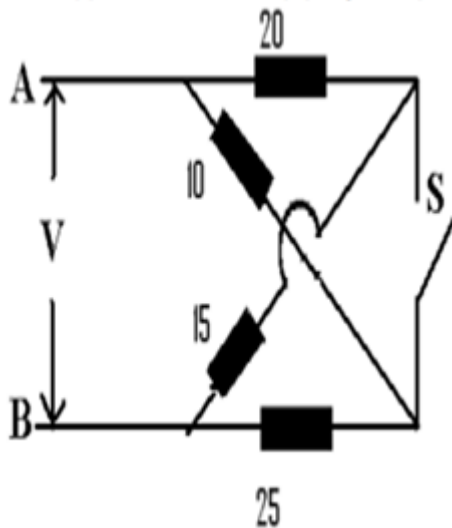
(i) Open

(ii) Closed

The current through 15Ω resistor is $3A$.

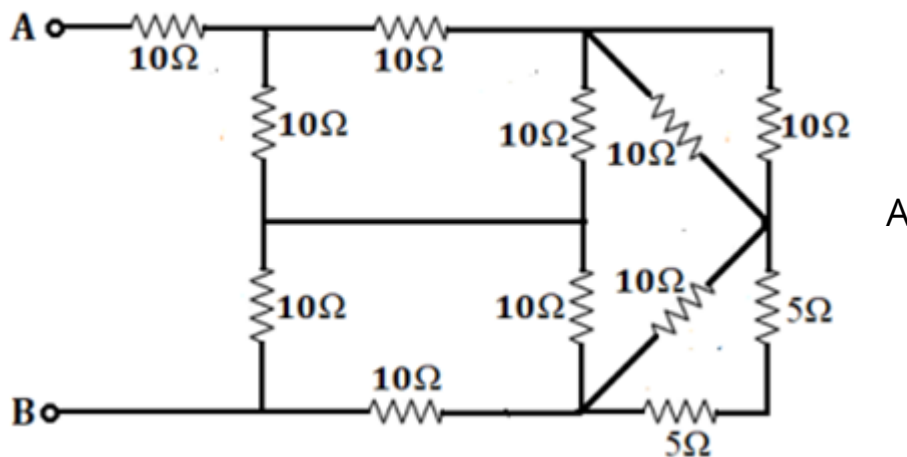
All the values of resistor is in ohms.

(6.0 Marks)



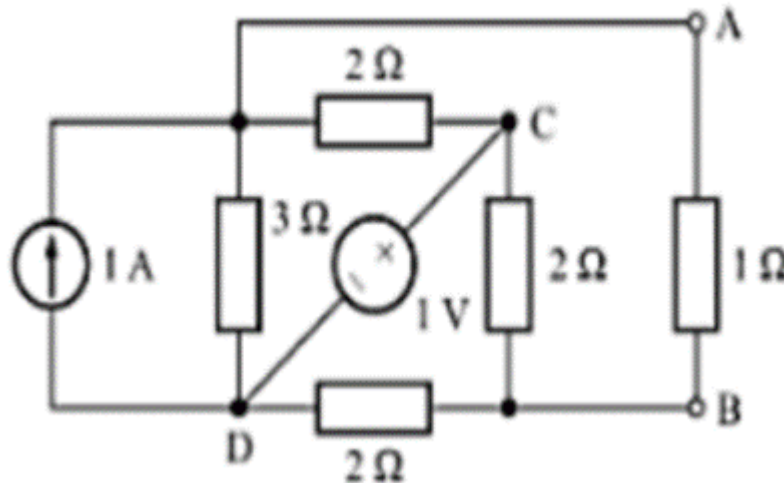
1.b. Find the equivalent resistance between the terminals A and B by employing star - Delta transformations , in the network shown:

(7.0 Marks)



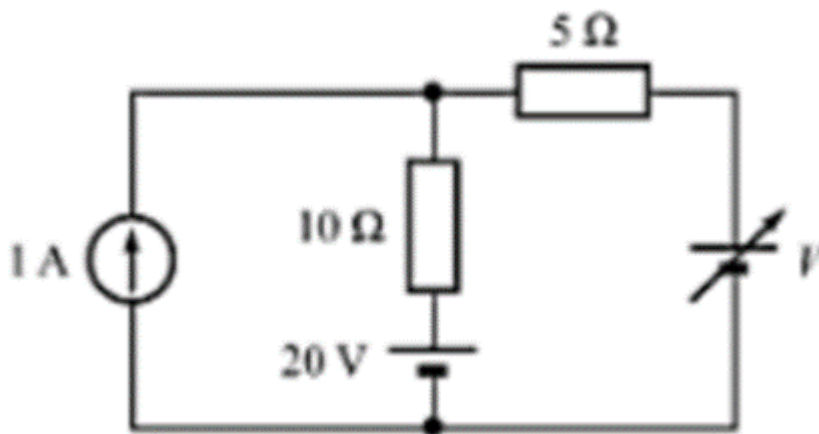
1.c. Using **Thevenin's theorem** , Find the current through 1Ω resistor in the circuit shown below :

(7.0 Marks)



1.d. In network shown below , Find the value of **V source** which results in Zero current in 5Ω resistor in clockwise wise direction .
Use **Superposition theorem only** .

(5.0 Marks)



2.a. An EMF whose instantaneous value at time t is given by $300\sin(100\pi t + \frac{\pi}{4})$ volts, is applied to an inductive circuit and the current in the circuit is given by $25\sin(100\pi t - \frac{\pi}{6})$ Amperes.

Find :

- (i) Frequency of applied Voltage .
- (ii) Resistance of the circuit
- (iii) Inductance of the circuit
- (iv) Active power drawn .
- (v) Value of series capacitance to be connected in order to get circuit into resonance
- (vi) Total impedance after adding capacitor in series for resonance

(7.0 Marks)

2.b. The admittance of a 2 element circuit is given by $(0.08 - 0.05i)$ mhos. Find the values of Resistance and Reactance of the circuit when elements are connected in :

(i) Series

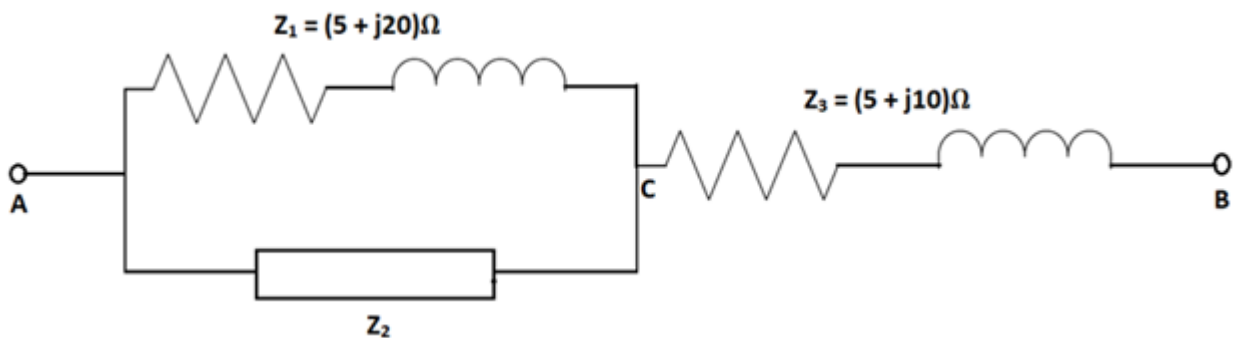
(ii) Parallel

(5.0 Marks)

2.c. When a 220V AC supply is applied across terminals A & B of the circuit shown, the total power input is 3.25kW and the total current is 20A, lag. Find :

(i) The complex expressions for currents through Z_1 and Z_2 , taking V_{AC} as reference phasor.

(ii) Draw phasor diagram



(7.0 Marks)

2.d.

A single-phase motor takes 8.3 A at a power factor of 0.66 lagging when connected to a 230 V, 50 Hz supply.

A Pure capacitor is now connected in parallel with the motor to raise the power factor to 0.9.

Determine the value of capacitor.

(6.0 Marks)

3.a. Write a short note on**i) Fuse****ii) Classification of cables based on Voltage rating** (5.0 Marks)

3.b. Three coils each having a resistance of 20Ω and reactance of 15Ω are connected in star, across a balanced three-phase, 400 V, 50-Hz supply. Calculate the readings on two wattmeters connected to measure the power input. (5.0 Marks)

3.c. With the help of a neat circuit diagram and phasor diagram , prove that two wattmeters are sufficient to measure total 3 phase active power. Also derive the expression for power factor of the circuit in terms of wattmeter readings. (7.0 Marks)

3.d. The list of loads and average consumption hours per day of a typical household is given below :

S.No.	Name of the Appliance	Wattage	Average consumption hours per day
1.	Four LED Bulbs	15 W each	14 hours each
2.	Four Ceiling fans	75 W each	12 hours each
3.	Geyser	1 kW	1 hour
4.	Refrigerator	100 W	24 hours
5.	Television	50 W	8 hours
6.	Mixer Grinder	750 W	15 minutes

Considering a 31 day month, Determine

i) Total number of units consumed in the month

ii) Monthly bill for the above consumption units considering a domestic connection of 4 kW sanctioned load with the tariff details given below :

S.No	Type of Charges	Tariff Details
1.	Fixed Charges for the sanctioned load	Rs. 100/- - For first kW Rs. 110/- - For every additional kW
2.	Energy consumption Charges	0 – 50 Units @ Rs. 4.15/- Per Unit 51 – 100 Units @ Rs. 5.6/- Per Unit 101 – 200 Units @ Rs. 7.15/- Per Unit Above 200 Units @ Rs. 8.2/- Per Unit
3.	Fuel Adjustment Charges	@ 8 Paise per unit consumed
4.	Tax applicable only for units consumed	@ 9% (Note: Only applicable on Energy consumption charges)

(8.0 Marks)

4.a. A 4 pole, 50 Hz three phase induction motor is running on a balanced three phase 400V, 50 Hz supply. Determine

i) Synchronous speed

ii) No Load speed if No Load slip is 1%

iii) Full Load slip if full load speed is 1440 rpm

iv) Frequency of rotor currents under Full Load condition

v) Rotor frequency when the motor is just about to start (5.0 Marks)

4.b. A 4 pole Separately excited DC Motor operating on a 200V DC supply has a wave wound armature with 500 conductors & armature resistance 0.5Ω . If the torque developed is 20 Nm & back EMF is 190V, determine

i) current drawn by the motor

ii) Useful flux per pole

iii) Speed of the motor

iv) Armature power developed (5.0 Marks)

4.c. i) With proper nomenclature, derive an expression for EMF generated in a separately excited DC Generator.

ii) Mention any two applications of BLDC Motor (7.0 Marks)

4.d. A 10kVA, 2000V/200V, 50 Hz single phase transformer has 100 secondary turns and a maximum flux density of 1.2 Tesla. On conducting OC and SC Tests, the wattmeter readings are found to be 100W & 400W respectively. Determine

- i) Rated Primary and Secondary Currents**
- ii) Net cross sectional area of the core**
- iii) Maximum efficiency at 0.8 Lag powerfactor**
- iv) Overall Maximum efficiency**

(8.0 Marks)